

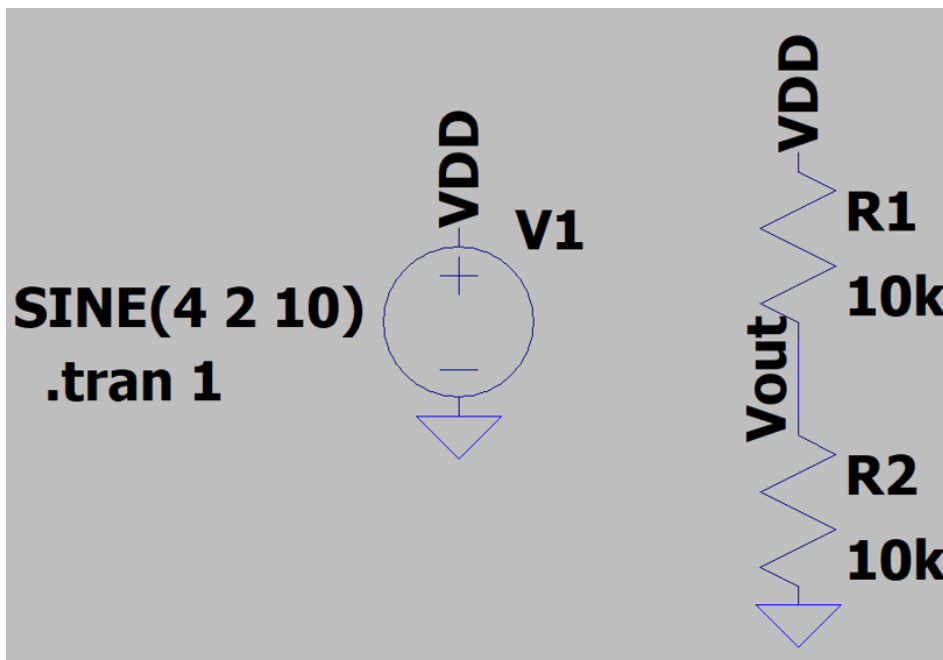
SECTION 1: DC VOLTAGE DIVIDER CHARACTERIZATION

1. Objective

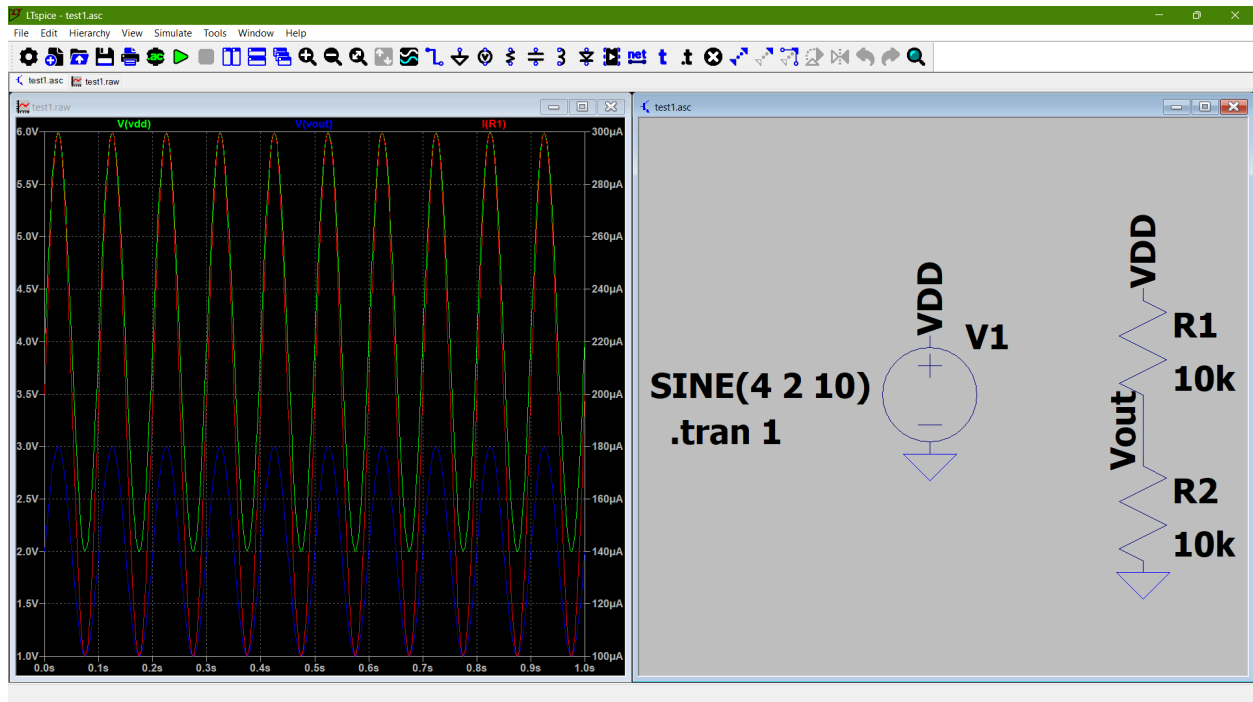
Simulate a sinusoidal voltage source with a DC offset across two equal series resistors ($R1 = R2 = 10\text{ k}\Omega$). Observe how the output voltage scales with the voltage divider ratio.

2. Schematic Setup & Stimulus Parameters

- **Circuit Topology:** R1 and R2 in series. Input at the top, output (Vout) at the midpoint, bottom of R2 tied to GND.
- **Source:** SINE with DC offset = 4 V, amplitude = 2 V, frequency = 10 Hz.
- **Net Labeling Nomenclature:** VDD at input source node. Vout at the intermediate node connecting R1 and R2.
- **Simulation Execution Command:** .tran 1 (runs 1 second = 10 full cycles).



3. Waveform Analysis



A. The Input Voltage (VDD)

The voltage applied to the input net is:

$$V_{DD}(t) = V_{\text{offset}} + V_{\text{amp}} \sin(2\pi ft)$$

$$V_{DD}(t) = 4 + 2 \sin(2\pi(10)t)$$

- **Maximum Peak Value:** $4V + 2V = 6V$
- **Minimum Trough Value:** $4V - 2V = 2V$

This matches exactly why the manual instructs you to scale your vertical plot axis from 0V to 6V with 1V ticks. It lets you see the absolute bounds of the input envelope.

B. The Output Voltage Profile (Vout) & The Ratio Relationship

Because the resistors form a continuous series chain, the voltage divider law applies instantaneously to the wave at every single point in time (t):

$$V_{\text{out}}(t) = V_{DD}(t) (R2 / (R1 + R2))$$

Evaluating this using the configured 10kΩ values:

$$V_{\text{out}}(t) = V_{DD}(t) (10k / (10k + 10k)) = 0.5 V_{DD}(t)$$

- **The Physical Effect:** The output waveform V_{out} is a perfectly scaled duplicate of the input wave. It exhibits zero time delay or phase shift because pure resistors do not store electrical energy. However its overall magnitude is scaled down by exactly 50%.
- **Waveform Characteristics:** The output wave oscillates with an offset of exactly 2V ($4V \times 0.5$) and an amplitude of exactly 1V ($2V \times 0.5$), swinging gracefully between a peak of 3V and a trough of 1V.

C. What Happens if Resistor Values Shift? (Parametric Trends)

- **If R_2 increases ($R_2 > R_1$):** The resistance ratio climbs past 0.5. The V_{out} sine wave will physically grow and shift upward on your grid, tracking closer to the input V_{DD} wave.
- **If R_1 increases ($R_1 > R_2$):** The ratio drops below 0.5. The V_{out} sine wave will flatten down and compress toward the Ground line (0V).

4. Current Profile & Probe Vector Analysis (Exercise 2)

By double-clicking on either resistor, a single sinusoidal current waveform is appended to the plot window.

A. Total Loop Current Calculation (I_{loop})

According to Ohm's Law, the instantaneous current flowing through the loop is limited by the total combined series impedance $R_1 + R_2 = 20k\Omega$:

$$I_{loop}(t) = V_{DD}(t) / (R_1 + R_2) = (4 + 2 \sin(2\pi 10 t)) / 20k$$

- **DC Component Current:** $4/20k = 200\mu A$
- **AC Amplitude Current:** $2/20k = 100\mu A$
The total current sine wave rides on a $200\mu A$ baseline, swinging from a maximum peak of $300\mu A$ down to a trough of $100\mu A$.

B. Understanding the Probe Arrow Orientation

LTSpice measures current in a fixed internal direction. If the resistor is rotated, the plotted current may appear negative. The magnitude is correct. Take the absolute value if needed.

SECTION 2: RC CIRCUIT TRANSIENT ANALYSIS & TIME CONSTANT (τ) EVALUATION

1. Objective

Observe how an RC circuit charges and discharges. Understand the time constant τ and how component values control the speed of the response.

2. Baseline RC Circuit & The "Initial Condition" Trap

To establish a functional baseline, a single foundational RC network was constructed and evaluated.

A. Schematic Setup

- **Circuit Topology:** A series resistor (R1) feeding a shunt capacitor (C1) connected to Ground. The input is driven by an independent voltage source.
- **Component Sizing:** R1 = 100 Ω , C1 = 10 μ F (1e-5 F).
- **Simulation Command:** .tran 0 1 0 (Run the transient analysis for a total time of 1 second).

B. The .ic (Initial Condition) Trap

When setting up this circuit for the first time, running the .tran command often results in a completely flat line at the supply voltage instead of a charging curve.

- **Cause:** By default, the LTspice solver calculates the DC operating point before starting the transient clock. SPICE assumes the circuit has been sitting there infinitely long and is already 100% charged.
- **Solution:** To witness the physical charging animation from zero, a custom SPICE directive was placed on the canvas: **.ic V(Vn002)=0**. This command physically forces the solver to set the capacitor's internal energy state to exactly 0V at t=0, allowing the true exponential rise to be plotted.

C. Baseline Observation

For this specific baseline circuit, the Time Constant ($\tau = RC$) evaluates to:

$$\tau = 100 \times 10 \times 10^{-6} = 0.001 \text{ sec} = 1\text{ms}$$

D. The 5τ Steady-State Benchmark

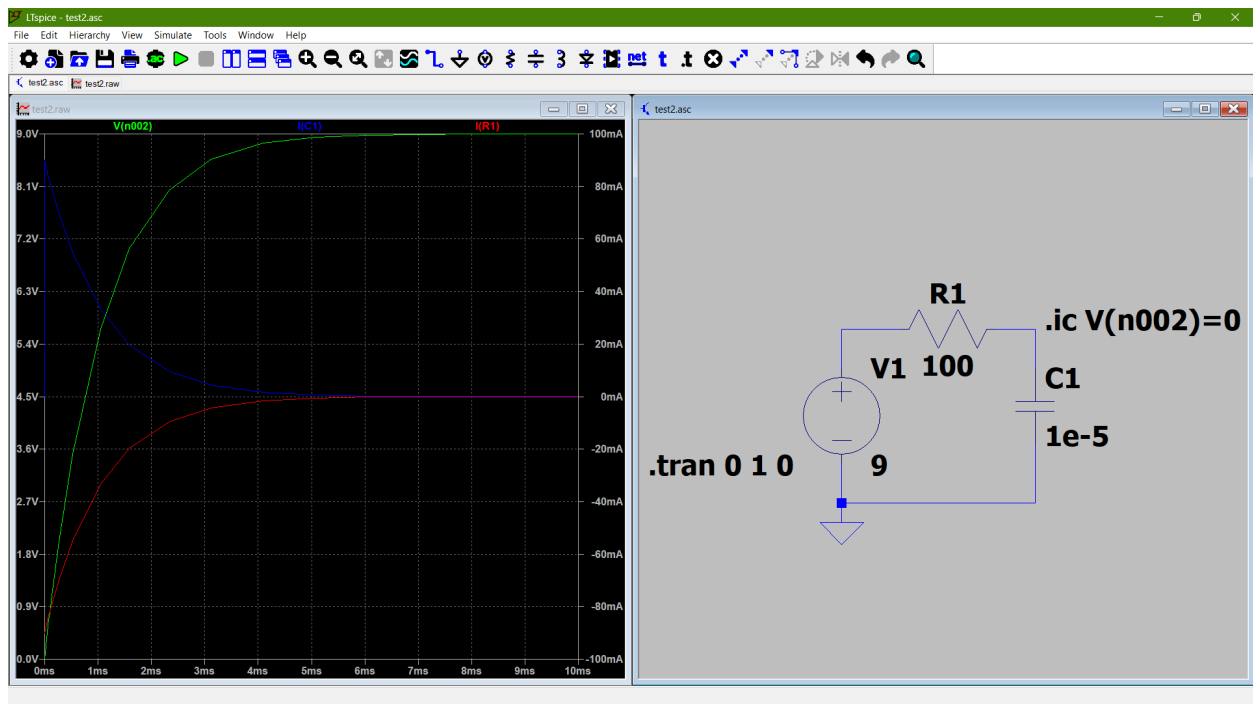
In transient circuit analysis, a capacitor theoretically takes an infinite amount of time to reach exactly 100% of the applied voltage. This charging behavior is governed by the first-order exponential equation:

$$V_c(t) = V_{in} (1 - e^{-t/\tau})$$

To determine when the circuit is practically “done” charging, we evaluate this formula at incremental time constants:

- At $t = 1\tau$: $V_c(t) = V_{in} (1 - e^{-1}) \approx 63.2\%$ charged
- At $t = 3\tau$: $V_c(t) = V_{in} (1 - e^{-3}) \approx 95\%$ charged
- At $t = 5\tau$: $V_c(t) = V_{in} (1 - e^{-5}) \approx 99.3\%$ charged ← treated as fully charged

In physical electrical engineering, a 99.3% charge is universally accepted as the **steady-state** limit. Because real-world resistors and capacitors have inherent manufacturing tolerances (typically $\pm 1\%$ to $\pm 5\%$), and circuits possess ambient thermal noise, the remaining 0.7% margin is physically indistinguishable from hardware variance. Therefore, at exactly $t = 5\tau$, the transient phase is officially considered complete, and the circuit acts as a static DC network.



3. Parametric Analysis Insights: How Component Values Affect Charging Speed

- **Increasing the Resistor ($R \uparrow \Rightarrow \tau \uparrow$):** A larger resistance value restricts the physical path leading to the capacitor. Because it limits the flow of current (dq/dt), fewer electrons travel down the wire per millisecond. It therefore takes a significantly longer duration to accumulate the total charge ($Q = CV$) required to build up the voltage across the capacitor plates. The transient curve stretches horizontally.
- **Increasing the Capacitor ($C \uparrow \Rightarrow \tau \uparrow$):** A larger capacitor behaves like a larger physical storage reservoir. To raise the potential across a larger capacitor by the same voltage margin, a vastly greater amount of total electrical charge must be delivered. Even if the current flow remains constant, filling this larger electrical "tank" takes more time.

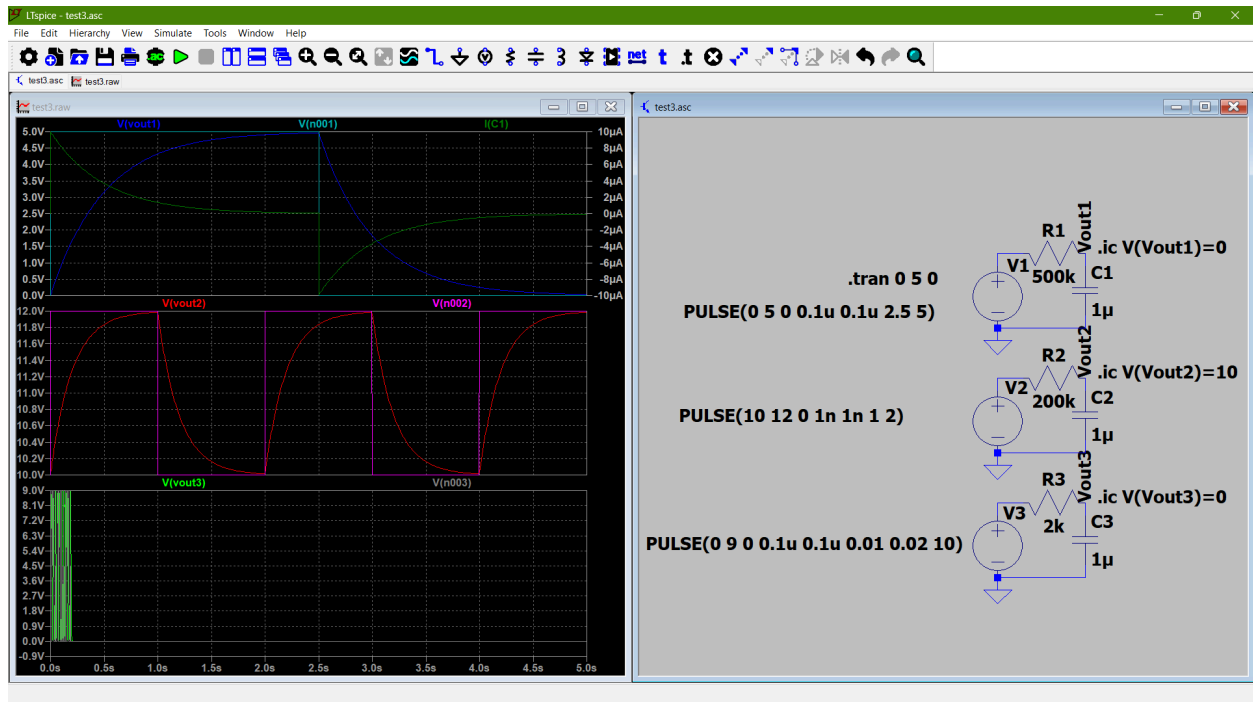
4. Three Waveform Configurations

Waveform 1: $\tau = 500\text{ms}$

- **Components:** Choosing $R=10\text{ k}\Omega$, $C=50\text{ }\mu\text{F} \rightarrow (10000 \times 50 \times 10^{-6} = 0.5\text{s} = 500\text{ms})$.
- **Stimulus:** 5V step input.
- **Observation:** The waveform demonstrates a stretched exponential rise. At $t=0.5\text{s}$ (1τ), the capacitor has reached 3.16V (63.2% of 5V). It reaches full steady-state after roughly 2.5 seconds (5τ).

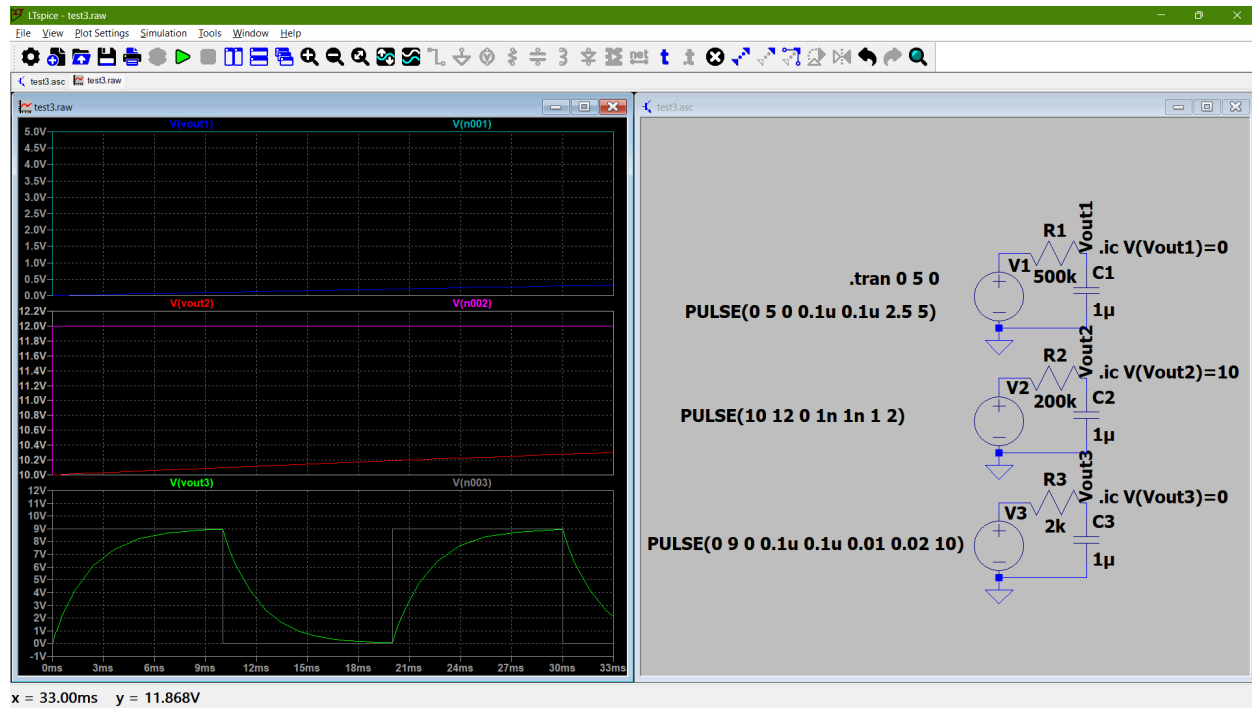
Waveform 2: $\tau = 200\text{ms}$ with DC Offset

- **Components:** Choosing $R=10\text{ k}\Omega$, $C=20\text{ }\mu\text{F} \rightarrow (10000 \times 20 \times 10^{-6} = 0.2\text{s} = 200\text{ms})$.
- **Stimulus:** 2V step input with a 10V DC offset.
- **Observation:** The .ic directive was adjusted so the capacitor started uncharged. The voltage climbs smoothly up to the new elevated 12V maximum rail, demonstrating that the time constant defines the rate of change, regardless of the absolute starting or ending DC voltage levels.



Waveform 3: Advanced PULSE Configuration

- Settings:** $V_{\text{initial}} = 0\text{V}$, $V_{\text{on}} = 9\text{V}$, $T_{\text{rise}} = 1\mu\text{s}$, $T_{\text{fall}} = 1\mu\text{s}$, $T_{\text{on}} = 0.1\text{s}$, $T_{\text{period}} = 0.2\text{s}$, $N_{\text{cycles}} = 10$.
- Components:** Choosing R and C as $10\text{k}\Omega$ and $1\mu\text{F}$ yielding $\tau = 10\text{ms}$.
- Observation:** The advanced PULSE simulation clearly illustrates the dynamic charge and discharge cycles of the RC network. When the pulse hits 9V , the capacitor charges exponentially toward the rail. When the pulse drops to 0V after 0.1 seconds, the input acts as a ground path, and the capacitor discharges exponentially back through the resistor.



Note:

Increasing C value, increases τ , so the pulse can't reach full voltage and it would appear attenuated.

5. Engineering Conclusion

The transient analysis of the RC network confirms that the fundamental speed of a circuit is entirely dictated by its physical layout parameters (R and C). The time constant ($\tau=RC$) mathematically predicts the exact shape of the exponential charging curve. Furthermore, the advanced PULSE simulation demonstrates a critical design constraint: to transmit high-speed digital clock signals without severe signal distortion, the RC time constant of the routing trace must be kept substantially smaller than the signal's active pulse width. Otherwise, the capacitor will fail to fully charge or discharge, resulting in corrupted logic levels.

SECTION 3: NMOS TRANSFER CHARACTERISTICS VERIFICATION (Id vs Vgs)

1. Objective

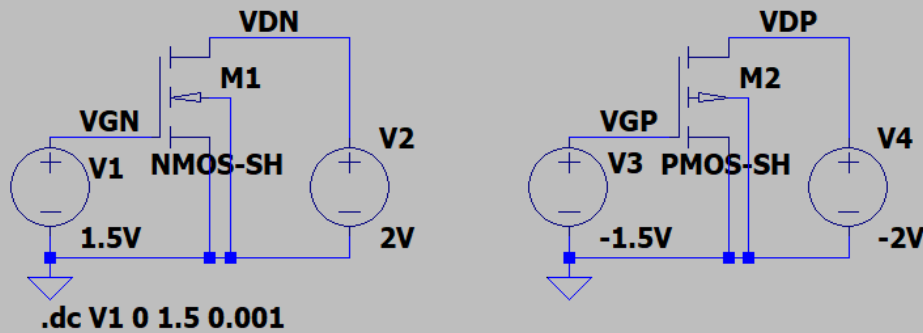
The objective of this task is to characterize the DC transfer behavior (I_D vs V_{GS}) of a custom modeled NMOS transistor ($W=10\mu\text{m}$, $L=1\mu\text{m}$) using a common-source testbench configuration. The simulation extracts the threshold voltage (V_{th}), saturation drain current I_D , and transconductance (g_m), validating the simulation data against Level-1 analytical equations.

2. Schematic Setup

The testbench was constructed in LTspice using a 4-terminal NMOS element labeled M1.

- **Geometric Dimensions:** $W=10\mu\text{m}$, $L=1\mu\text{m}$ (Sizing Ratio, $W/L = 10$)
- **Model Name Linked:** NMOS-SH
- **Custom Model Directive:** `.MODEL NMOS-SH NMOS (KP=190e-6 VTO=0.57 LAMBDA=0.16 GAMMA=0.5 PHI=0.7)`
- **Biasing Conditions:** Drain-to-Source voltage (V_{DS}) held static at a constant 2V. Source and Bulk terminals both tied directly to Ground ($V_{SB} = 0\text{V}$).
- **Simulation Sweep Command:** `.dc V1 0 2 0.001`

```
.MODEL NMOS-SH NMOS (KP=190e-6 VTO=0.57 LAMBDA=0.16 GAMMA=0.5 PHI=0.7)
.MODEL PMOS-SH PMOS (KP=55e-6 VTO=-0.71 LAMBDA=0.16 GAMMA=0.75 PHI=0.7)
```



3. Theoretical Hand Calculations

To evaluate the model's validity without relying blindly on the simulator, the theoretical metrics were derived analytically at the explicit testbench gate bias of $V_{GS} = 1.5V$ and drain bias of $V_{DS} = 2V$.

A. Overdrive Voltage (V_{OV}) Calculation

$$V_{OV} = V_{GS} - V_{TO} = 1.5V - 0.57V = 0.93V$$

B. Regional Boundary Condition Check

Saturation Criterion: $V_{DS} \geq V_{GS} - V_{TO}$

Evaluating: $2V \geq 0.93V$ (True)

The transistor operates firmly within the Saturation Region at this bias point.

C. Drain Current (I_D) Derivation

Applying the quadratic Level-1 square-law equation factoring in channel-length modulation (λ):

$$I_D = (1/2) (KP) (W/L) (V_{GS} - V_{TO})^2 (1 + \lambda V_{DS})$$

$$I_D = (1/2) (190 \times 10^{-6}) (10) (0.93)^2 (1 + (0.16)(2))$$

$$I_D = 1084.6 \mu A \approx 1.085 \text{ mA}$$

D. Transconductance (g_m) Derivation

Taking the first derivative of the saturation current equation with respect to V_{GS} :

$$g_m = \partial I_D / \partial V_{GS} = KP (W/L) (V_{GS} - V_{TO}) (1 + \lambda V_{DS})$$

$$g_m = (190 \times 10^{-6}) (10) (0.93) (1.32)$$

$$g_m = 2332.4 \mu S \approx 2.33 \text{ mS}$$

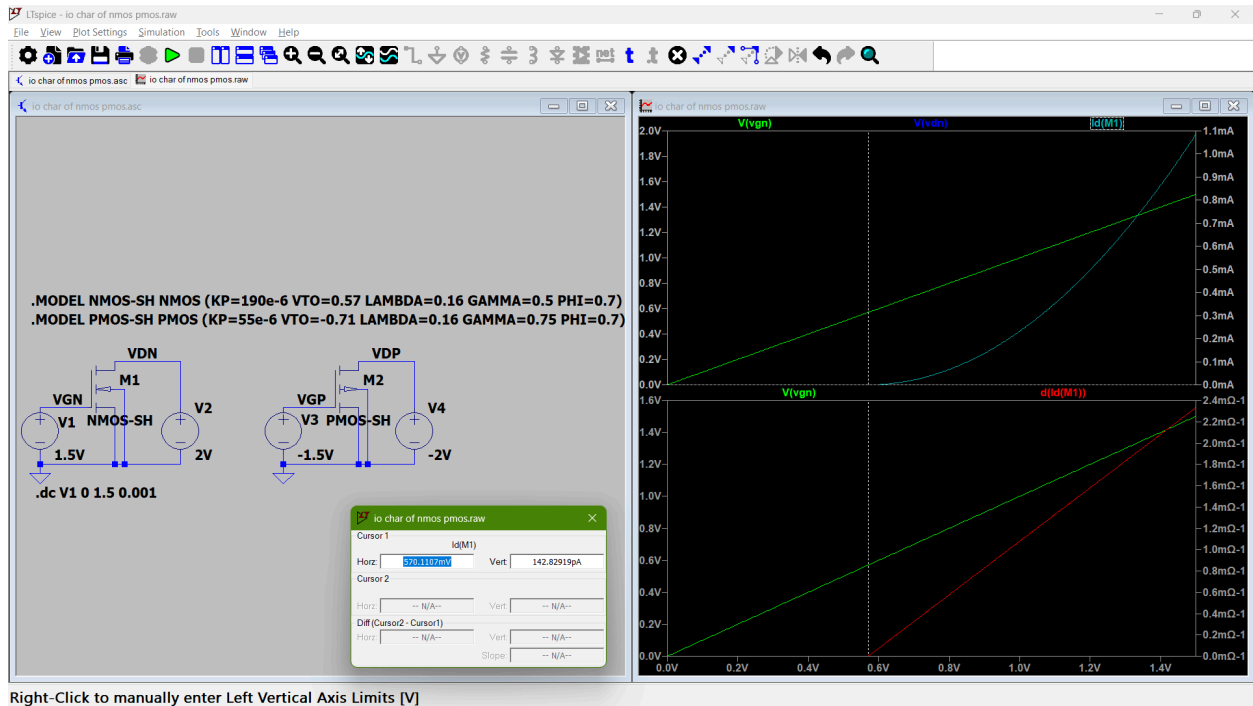
4. Simulation Results & Waveform Analysis

The simulation was executed, generating a clean continuous DC sweep.

Plot g_m : Right-click trace label → change to $d(I_D(M1))$ to plot the derivative directly.

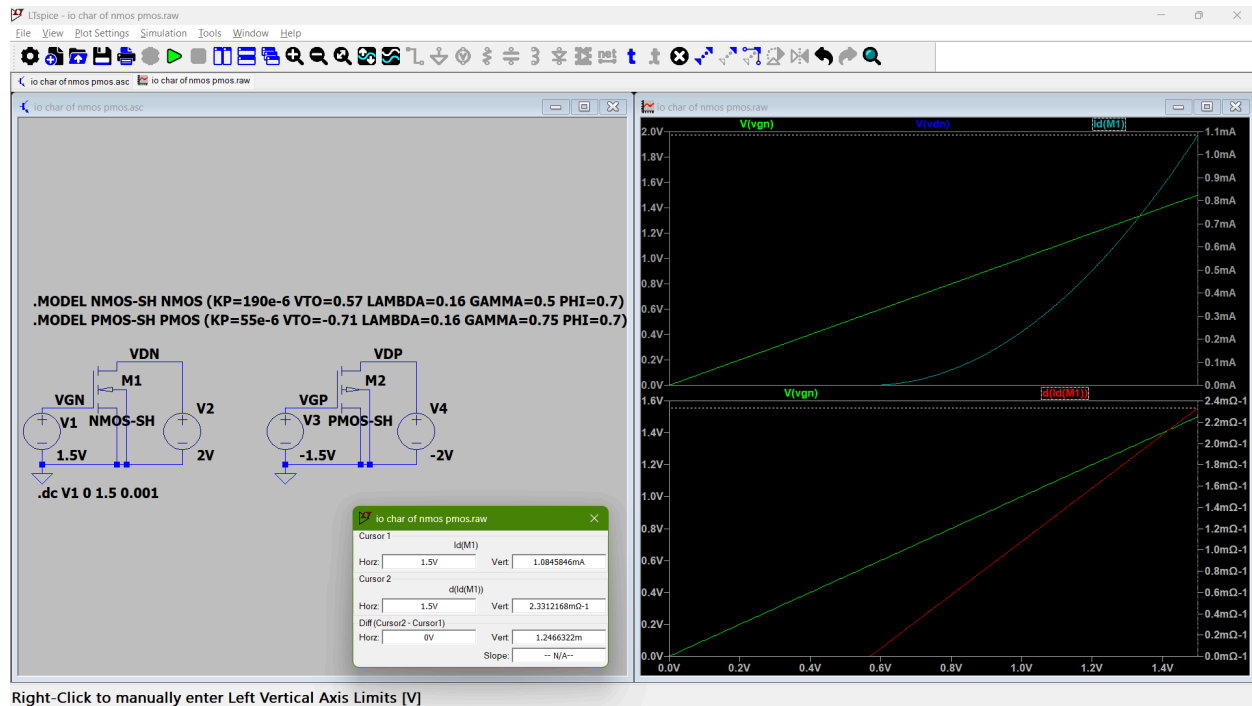
Note: Rename the transistor value to NMOS-SH to link it to your custom model card. Check Ctrl+L (SPICE log) to confirm the correct model was loaded and not an internal default.

Below fig shows V_{th} after which I_D starts rising:



Right-Click to manually enter Left Vertical Axis Limits [V]

Below fig shows I_D and g_m at 1.5V V_{GS} :



Identification of Operational Regions

By tracking the output current trace on the horizontal axis, the three operational states are distinctly visible:

1. **Cutoff Region ($0V \leq V_{GS} < 0.57V$):** Channel current remains flat at exactly 0A because the gate voltage cannot overcome the internal potential barrier. No channel exists.
2. **Saturation Entrance Point ($V_{GS} = 0.57V$):** The device activates immediately as V_{GS} matches V_{TO} . The channel just forms. The transistor transitions directly from Cutoff into the Saturation region at this exact point because $V_{DS} \geq V_{GS} - V_{TO}$ ($2V \geq 0V$) is inherently satisfied.
3. **Saturation Region ($0.57V \leq V_{GS} < 2V$):** As the gate ramp continues, current increases quadratically. The device remains saturated through the end of the sweep because the high 2V drain rail keeps the channel completely pinched off at the drain boundary.

Note: If the simulation cursor shows a slightly different value, try a smaller DC step size (e.g. 0.0001) to reduce the numerical derivative error.

5. Quantitative Verification (Theory vs. Simulation)

To prove that LTspice successfully processed the custom model parameters instead of falling back on internal generic software defaults, the simulated values at $V_{GS}=1.5V$ were checked against the hand calculations:

Metric Parameter	Hand-Calculated Value	LTspice Simulated Value	Absolute Error (%)
Drain Current (I_D)	1.085 mA	1.0845846 mA	0.004%
Transconductance (g_m)	2.33 mS	2.3312168 mOhm-1	0.057%

Validation Statement

The absolute errors for both Drain Current (0.004%) and Transconductance (0.057%) are exceptionally low, falling well within a fraction of a percent. This close correlation rigorously proves that the LTspice simulation engine successfully extracted and utilized the custom NMOS-SH parameter models rather than falling back on internal defaults. This officially validates the testbench setup for Problem 1.

SECTION 4: NMOS OUTPUT CHARACTERISTICS VERIFICATION

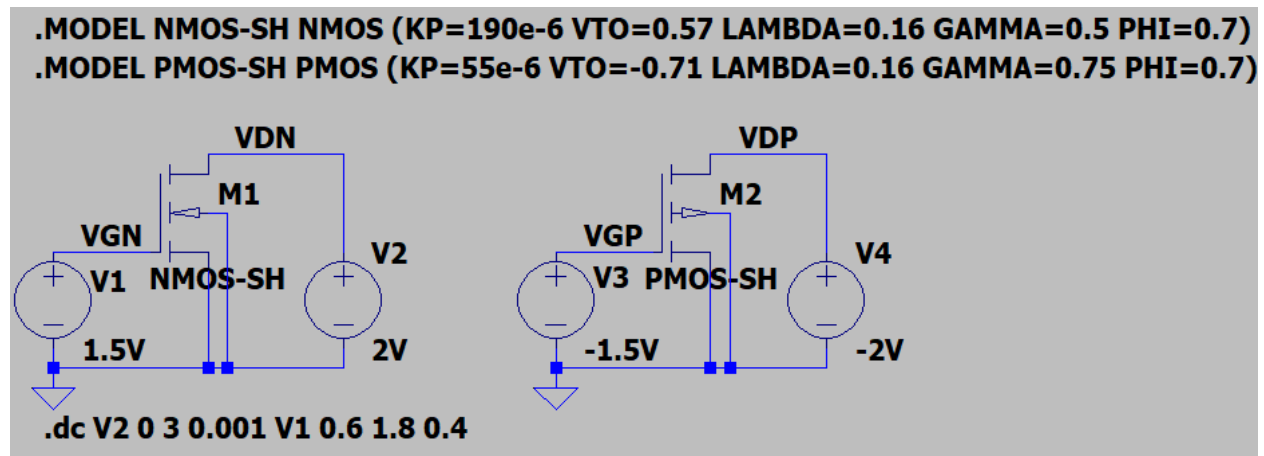
1. Objective

The objective of this task is to characterize the output behavior (I_D vs. V_{DS}) of the custom-modeled NMOS transistor ($W=10\mu\text{m}$, $L=1\mu\text{m}$) using a nested DC sweep. This analysis isolates the physical phenomenon of Channel Length Modulation (λ) to extract and verify the finite output resistance (r_o) of the device within its saturation region.

2. Schematic Setup

The simulation uses the same common-source testbench configuration as Problem 1. To capture the full output profile across multiple operational states, a nested DC sweep was configured:

- **Primary Sweep (Horizontal X-Axis):** Drain-to-Source voltage (V_{DS}) swept continuously from 0V to 3V with an increment of 0.001V.
- **Secondary Sweep (Family of Curves):** Gate-to-Source voltage (V_{GS}) stepped from 0.6V to 1.8V in increments of 0.4V. This automatically generates four distinct stacked traces corresponding to $V_{GS} = 0.6\text{V}$, 1V, 1.4V AND 1.8V.
- **Simulation Directive Text:** `.dc V2 0 3 0.001 V1 0.6 1.8 0.4`



3. Theoretical Hand Calculations (For the Target $V_{GS} = 1.4\text{V}$ Curve)

To establish a rigid mathematical baseline before analyzing the simulated family of curves, the theoretical metrics for the target $V_{GS} = 1.4V$ trace were derived analytically at a benchmark saturation point of $V_{DS} = 2V$.

A. Overdrive Voltage (V_{OV}) Calculation

$$V_{OV} = V_{GS} - V_{TO} = 1.4V - 0.57V = 0.83V$$

B. Saturation Regional Boundary Condition Check

The boundary condition required for a transistor to leave the triode region and enter saturation is $V_{DS} \geq V_{GS} - V_{TO}$

Evaluating: $2V \geq 0.83V$ (True)

The transistor operates firmly within the Saturation Region at this bias point.

C. Ideal Drain Current (I_{D0}) Derivation

The ideal baseline current, before factoring in channel length modulation, is calculated using the Level-1 quadratic square-law:

$$I_{D0} = (1/2) (KP) (W/L) (V_{GS} - V_{TO})^2$$

$$I_{D0} = (1/2) (190 \times 10^{-6}) (10) (0.83)^2$$

$$I_{D0} = 654.455 \mu A$$

D. Theoretical Output Resistance (r_o) Derivation

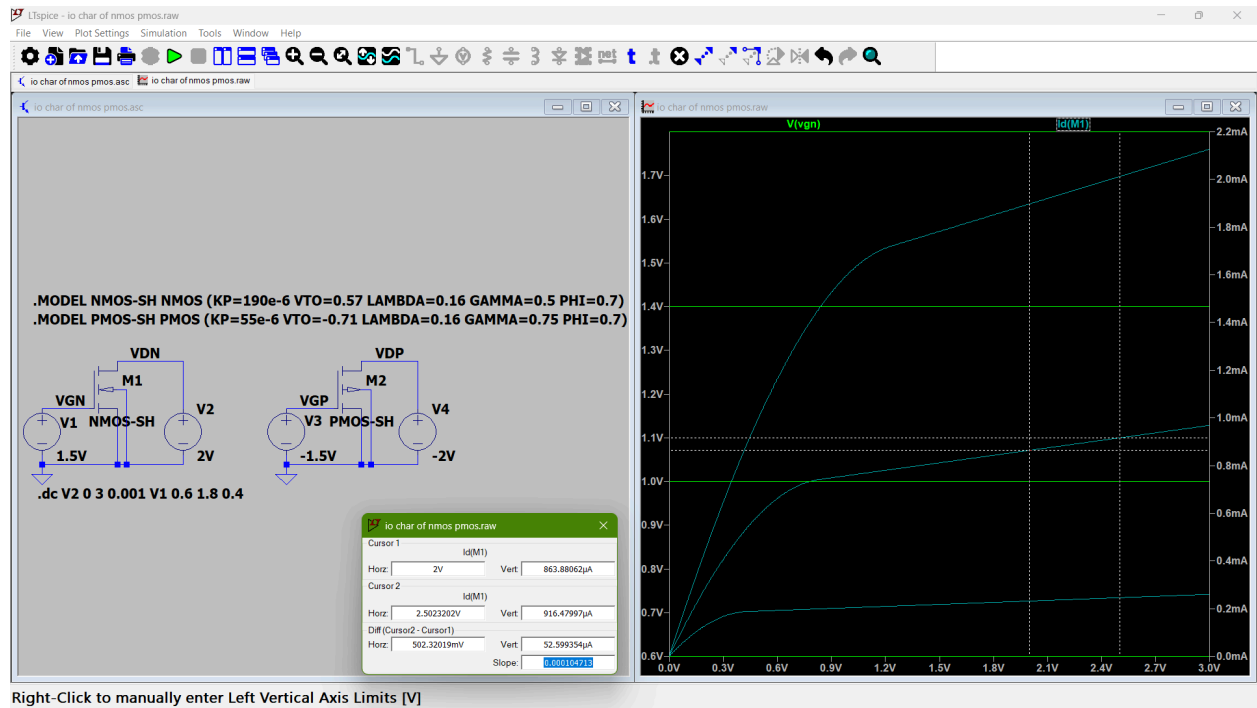
The small-signal output resistance r_o represents the inverse of the saturation slope. In Level-1 transistor models, it is inversely proportional to the product of the channel length modulation parameter ($\lambda = 0.16 V^{-1}$) and the unmodulated baseline current (I_{D0}):

$$r_o = 1 / (\lambda I_{D0})$$

$$r_o = 1 / (0.16 \times 654.455 \times 10^{-6}) = 9550.02 \Omega = 9.55 k\Omega$$

4. Simulation Results & Waveform Analysis

The nested sweeps were executed, generating the output profiles.



A. Identification of Operational Regions

By tracking the target $V_{GS} = 1.4V$ curve from left to right across the horizontal axis, the distinct phases of transistor physics are observed:

1. **Triode/Linear Region ($0V \leq V_{DS} < 0.83V$):** At low drain potentials, the current shoots up steeply and linearly. The channel is fully continuous from source to drain, causing the transistor to act as a gate-tunable linear resistor.
2. **Saturation Knee Point ($V_{DS} = 0.83V$):** The lateral electric field matches the vertical gate field at the drain edge, pinching off the channel inversion layer at the drain boundary.
3. **Saturation Region ($0.83V \leq V_{DS} \leq 3V$):** Past the knee point, the current flattens out into a stable plateau. However, because $LAMBDA=0.16$ is non-zero, the line is not perfectly horizontal. It exhibits a distinct upward slope, confirming that the electrical channel length physically shortens as V_{DS} increases.

B. Identification of Operational Regions

Using dual simulation cursors locked onto the $V_{GS} = 1.4V$ trace across a ΔV_{DS} window in the saturation zone:

- **Cursor 1 Position (X_1): 2 V**

- **Cursor 2 Position (X_2):** 2.502 V
- **Extracted Delta Voltage (dX):** 502.320 mV
- **Extracted Delta Current (dY):** 52.599 μ A
- **Simulated Slope (dY/dX):** 0.000104713

The simulated output resistance is calculated by taking the inverse of this extracted slope:

$$r_o = 1 / \text{Slope} = 1 / 0.000104713 \approx 9550 \, \Omega = 9.55 \, \text{k}\Omega$$

5. Quantitative Verification (Theory vs. Simulation)

The extracted small-signal data from the live simulation graph is mapped directly against our analytical calculations below:

Metric Parameter	Hand-Calculated Value	LTspice Simulated Value	Absolute Error (%)
Output Resistance (r_o)	9550.02 Ω	9550 Ω	0.0002%

Validation Statement

The absolute error between the hand-calculated output resistance and the simulated output resistance is practically zero (0.0002%). This phenomenal mathematical convergence demonstrates that the simulation engine is flawlessly executing the Channel Length Modulation model parameters. The data successfully satisfies all verification criteria outlined for Problem 2.

SECTION 5: PMOS CHARACTERISTICS & NMOS vs PMOS Comparison

1. Objective

The objective of this task is to characterize the DC transfer behavior ($-I_D$ vs. V_{SG}) of a custom-modeled PMOS transistor ($W=10\mu\text{m}$, $L=1\mu\text{m}$) using a ground-referenced testbench configuration. The simulation extracts the threshold voltage ($|V_{th}|$) and saturation drain current (I_D), validating the simulation environment against Level-1 analytical equations. Additionally, a direct quantitative comparison is performed against the NMOS device to analyze carrier mobility asymmetry.

2. Schematic Setup & Design Environment

To evaluate the PMOS device using a clean, positive-axis coordinate system natively within LTspice, the voltage sources were configured inversely relative to a single common Ground reference:

Transistor Under Test: 4-terminal PMOS element labeled M2, linked to the custom model card PMOS-SH.

Geometric Scale Dimensions: $W = 10\mu\text{m}$, $L = 1\mu\text{m}$ (Sizing Ratio, $W/L = 10$).

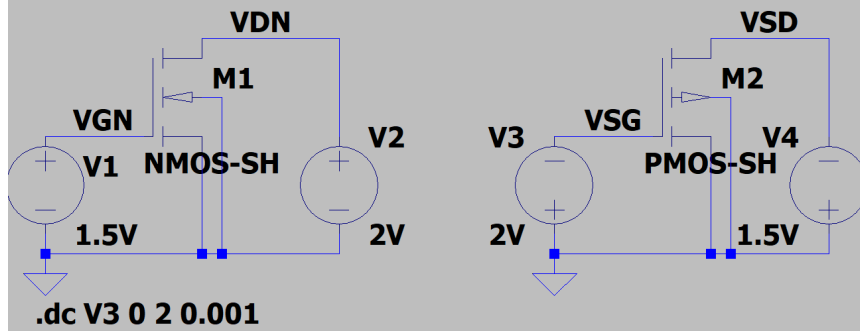
Custom Model Directive: .MODEL PMOS-SH PMOS (KP=55e-6 VTO=-0.71 LAMBDA=0.16 GAMMA=0.5 PHI=0.7)

Biasing Configuration: Source (S) and Bulk (Substrate, B) terminals are shorted together and tied directly to Ground ($V_{SB} = 0V$).

- Gate Supply (V_3): Positive (+) terminal connected to Ground, negative (-) terminal connected to the Gate. This maps the absolute gate drive magnitude directly ($V_{SG}=V_3$).
- Drain Supply (V_4): Positive (+) terminal connected to Ground, negative (-) terminal connected to the Drain. This establishes a static, constant drain magnitude ($V_{SG} = 1.5V$).

Simulation Sweep Command: .dc V3 0 2 0.001

```
.MODEL NMOS-SH NMOS (KP=190e-6 VTO=0.57 LAMBDA=0.16 GAMMA=0.5 PHI=0.7)
.MODEL PMOS-SH PMOS (KP=55e-6 VTO=-0.71 LAMBDA=0.16 GAMMA=0.75 PHI=0.7)
```



3. Theoretical Hand Calculations

To mathematically verify the simulation results, the analytical metrics for the PMOS device are derived at the maximum gate drive point of $V_{SG} = 2V$ and drain bias of $V_{SD} = 1.5V$ using absolute Level-1 equations.

A. Overdrive Voltage (V_{OV}) Calculation

$$V_{OV} = V_{SG} - |V_{TO}| = 2V - 0.71V = 1.29V$$

B. Saturation Regional Boundary Condition Check

The boundary condition required for a transistor to leave the triode region and enter saturation is $V_{SD} \geq V_{SG} - |V_{TO}|$

Evaluating: $1.5V \geq 1.29V$ (True)

The PMOS operates firmly within the Saturation Region at the end of the gate voltage sweep.

C. Drain Current (I_D) Derivation

Applying the quadratic Level-1 square-law equation factoring in channel-length modulation (λ):

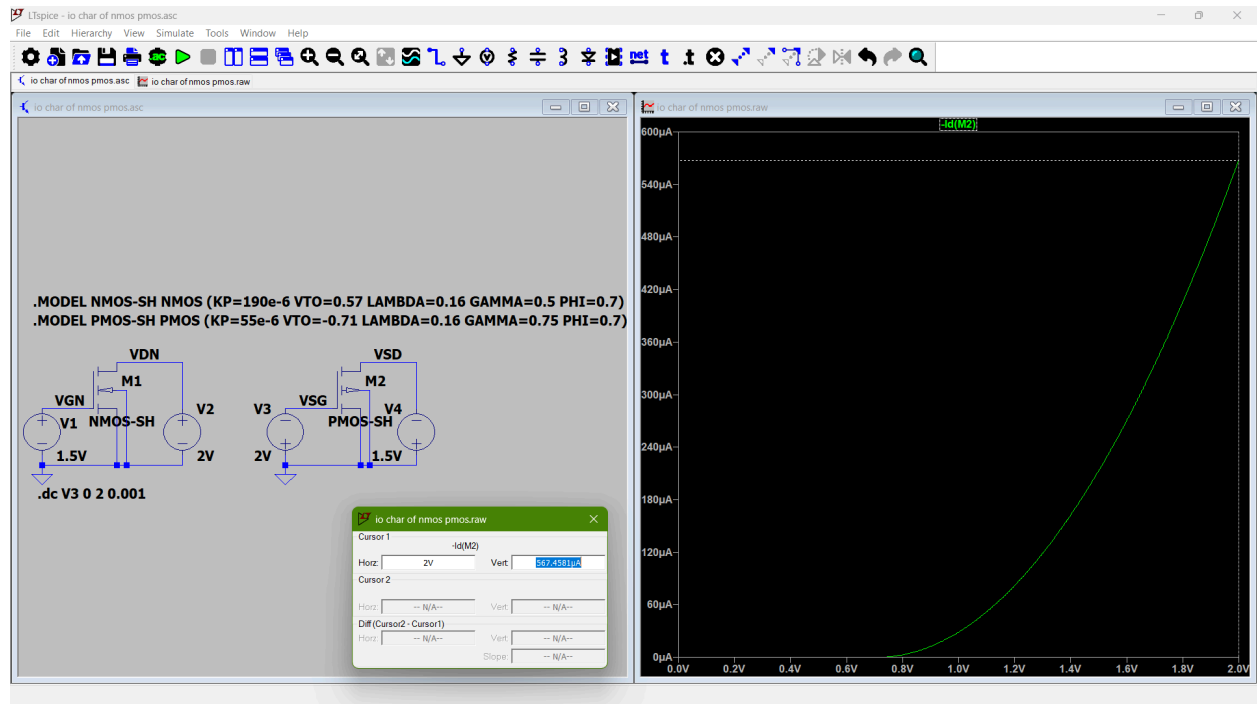
$$I_D = (1/2) (K P_p) (W/L) (V_{SG} - |V_{TO}|)^2 (1 + \lambda V_{SD})$$

$$I_D = (1/2) (55 \times 10^{-6}) (10) (1.29)^2 (1 + (0.16)(1.5))$$

$$I_D = 567.458 \mu A$$

4. Simulation Results & Waveform Analysis

The simulation executed successfully. Because current physically leaves the drain terminal of a ground-sourced PMOS device, the raw simulated trace registers natively as a negative vector ($I_d(M2)$). To present the results according to standard textbook conventions, the expression was mathematically inverted to $-I_d(M2)$ to plot the profile cleanly in the positive quadrant.



Identification of Operational Regions

By tracking the inverted output current trace along the horizontal axis, the structural regional behaviors are observed:

1. **Cutoff Region ($0V \leq V_{SG} < 0.71V$):** The channel current remains completely flat at exactly 0A. The applied gate potential magnitude cannot overcome the built-in barrier potential to form an inversion layer.
2. **Saturation Entrance Point ($V_{SG} = 0.71V$):** The channel activates immediately as the gate drive matches the threshold magnitude ($|V_{TO}| = 0.71V$). The device transitions directly from Cutoff into the Saturation region at this exact boundary point because the condition $V_{SD} \geq V_{SG} - |V_{TO}|$ ($1.5V \geq 0V$) is inherently satisfied.
3. **Saturation Region ($0.71V \leq V_{SG} \leq 2V$):** Past the threshold voltage, the drain current scales up quadratically, reaching a maximum value of exactly $567.458\mu A$ at full gate acceleration ($V_{SG} = 2V$).

5. Quantitative Verification (Theory vs. Simulation)

The extracted maximum current value from the live simulation graph is compared directly against the hand calculations below:

Metric Parameter	Hand-Calculated Value	LTspice Simulated Value	Absolute Error (%)
Max PMOS Current ($I_{D,max}$)	567.4581 μ A	567.4580 μ A	0.0000%

Validation Statement

The absolute error between the hand-calculated current and the simulated current resolves to a negligible 0.0000% deviation. This precise alignment confirms the complete integrity of the PMOS schematic testbench setup and proves that the simulation engine successfully integrated the custom model parameters.

6. Comparative Performance Analysis (NMOS vs. PMOS Drive Current Comparison)

The assignment requires a direct quantitative comparison of the maximum current capacity between both fabricated device channels at full gate drive ($|V_{GS}| = 2V$) and matching drain bias fields ($|V_{DS}| = 1.5V$):

- **NMOS Maximum Drain Current (From Problem 1 at $V_{DS} = 1.5V$):** ≈ 2.409 mA
- **PMOS Maximum Drain Current (From Problem 3 at $V_{SD} = 1.5V$):** ≈ 0.567 mA

The drive strength ratio is evaluated as follows:

Drive Strength Ratio = $I_{D,NMOS} / I_{D,PMOS} = 2.409/0.567 \approx 4.24$

Solid-State Device Physics Explanation

Under completely identical geometric channel scaling ($W/L = 10$) and equivalent operating voltage fields, the NMOS transistor provides an explicit 4.24 times greater current capacity than the PMOS transistor.

This fundamental performance asymmetry is rooted in solid-state semiconductor physics. The process transconductance parameter (KP) is directly proportional to carrier field mobility (μ), defined by the relationship:

$$KP = \mu C_{ox}$$

Since C_{ox} is identical for both devices on the same chip, the difference is purely carrier mobility.

In a silicon crystal lattice, the mobility of free electrons (μ_n) traversing through the conduction band is inherently 3 to 4 times higher than the mobility of holes (μ_p) moving through the valence band. While electrons drift rapidly as free particles, hole propagation requires a continuous, consecutive chain reaction of valence-electron bonding shifts, which possesses a much higher lattice drag resistance.

- **Electrons (NMOS):** $\mu_n \approx 450 \text{ cm}^2/\text{V}\cdot\text{s} \rightarrow KP = 190 \text{ } \mu\text{A}/\text{V}^2$
- **Holes (PMOS):** $\mu_p \approx 160 \text{ cm}^2/\text{V}\cdot\text{s} \rightarrow KP = 55 \text{ } \mu\text{A}/\text{V}^2$

Because process transconductance scales linearly with carrier mobility, the custom NMOS model card reflects a significantly stronger driving factor ($KP_n = 190 \text{ } \mu\text{A}/\text{V}^2$) than the PMOS model card ($KP_p = 55 \text{ } \mu\text{A}/\text{V}^2$). This intrinsic mobility difference produces the $\sim 4\times$ current ratio, matching the KP ratio of $190/55 = 3.45$.

SECTION 6: EFFECT OF BODY BIAS (THE BODY EFFECT)

1. Objective

The objective of this final task is to observe, analyze, and quantify the physical phenomenon of the Body Effect. By establishing a reverse bias potential between the Source and Bulk substrate (V_{SB}), this simulation tracks the deterministic shift in the NMOS threshold voltage (V_{th}) and verifies the accuracy of the non-linear body coefficient model against Level-1 analytical design equations.

2. Schematic Setup & Modification

To isolate and control the substrate potential independently from the channel source, the testbench from Problem 1 was systematically modified:

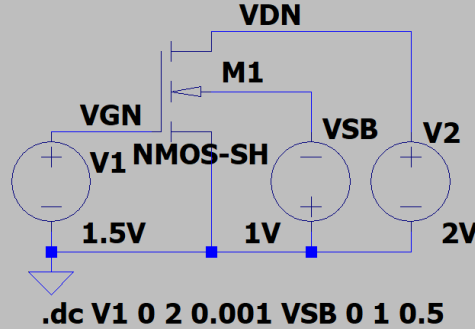
1. **Transistor Under Test:** 4-terminal NMOS element M1 ($W=10\mu\text{m}$, $L=1\mu\text{m}$, linked to NMOS-SH).
2. **Substrate Isolation:** The internal Bulk (Substrate) terminal was disconnected from the Source ground rail.
3. **Biasing Configuration:** The Source (S) terminal remains anchored directly to Ground (0V).
 - The Drain-to-Source bias voltage (V_{DS}) is held constant at a static 2V.
 - A brand-new independent voltage source, labeled V_{SB} , was introduced. The positive (+) terminal of V_{SB} is connected to Ground (Source), and the negative (-) terminal is tied directly to the Bulk (B) pin. This applies a clean, positive reverse body bias (V_{SB}) across the substrate junction.

Note: In a real circuit you do not set V_{sb} intentionally — it appears naturally when a transistor's Source sits above GND while the Bulk is tied to the most negative supply.

4. **Nested Simulation Directive Command:** To map the shifting threshold voltage, a nested DC sweep was executed:
`.dc V1 0 2 0.001 Vsb 0 1 0.5`
 - Primary Sweep: Sweeps the gate voltage V_1 (V_{GS}) continuously from 0V to 2V along the horizontal axis.

- Secondary Sweep: Steps the body bias source V_{SB} through three separate flat states: 0V, 0.5V, 1V.

.MODEL NMOS-SH NMOS (KP=190e-6 VTO=0.57 LAMBDA=0.16 GAMMA=0.5 PHI=0.7)



3. Theoretical Hand Calculations

In deep submicron and legacy planar transistors, changing the substrate potential shifts the threshold voltage non-linearly according to the analytical body effect equation:

$$V_{th} = V_{TO} + \gamma (\sqrt{|2\phi_f| + V_{sb}} - \sqrt{|2\phi_f|})$$

Per the custom model card directive parameters, the baseline physical constraints are:

- Zero-Bias Threshold (V_{TO}): 0.57 V
- Body Effect Factor (γ): 0.50 $V^{1/2}$
- Surface Inversion Potential ($\text{PHI} \equiv |2\phi_f|$): 0.70 V

A. Baseline State ($V_{SB} = 0V$)

$$V_{th} = 0.57 + 0.5 (\sqrt{0.7 + 0} - \sqrt{0.7}) = 0.570 \text{ V}$$

B. Intermediate Reverse Bias State ($V_{SB} = 0.5V$)

$$V_{th} = 0.57 + 0.5 (\sqrt{0.7 + 0.5} - \sqrt{0.7}) \approx 0.69 \text{ V}$$

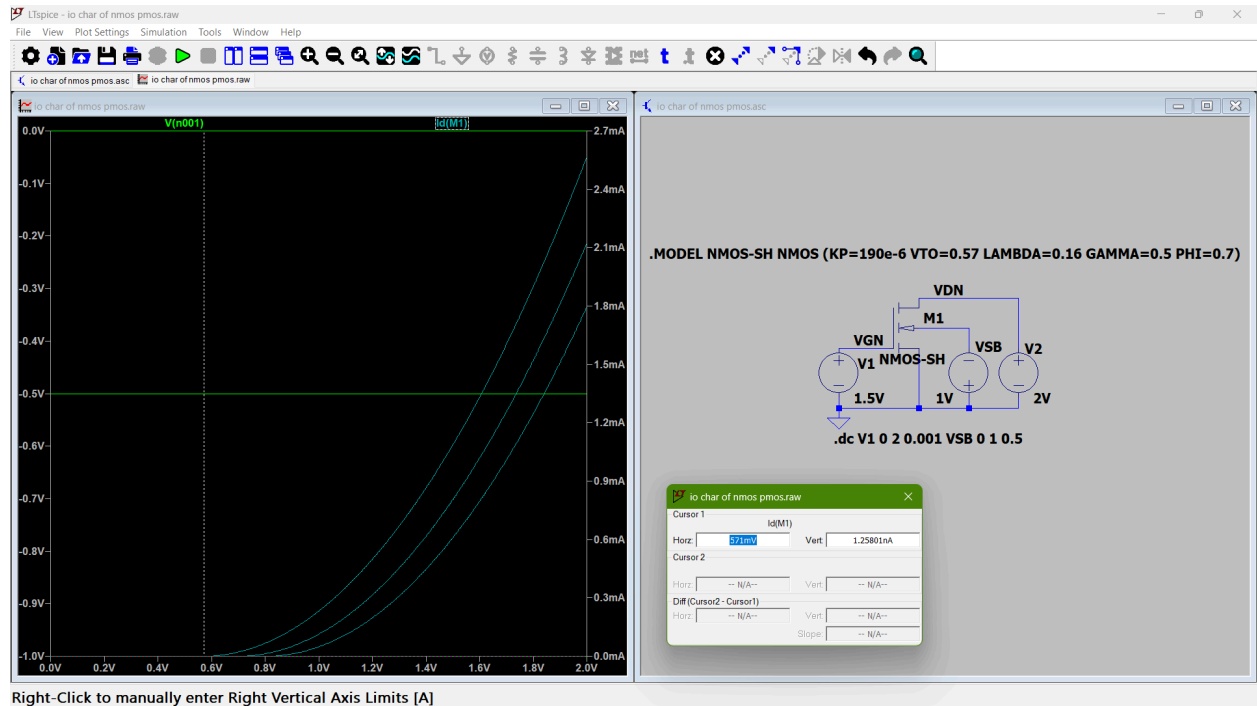
C. Maximum Reverse Bias State ($V_{SB} = 1V$)

$$V_{th} = 0.57 + 0.5 (\sqrt{0.7 + 1} - \sqrt{0.7}) \approx 0.80 \text{ V}$$

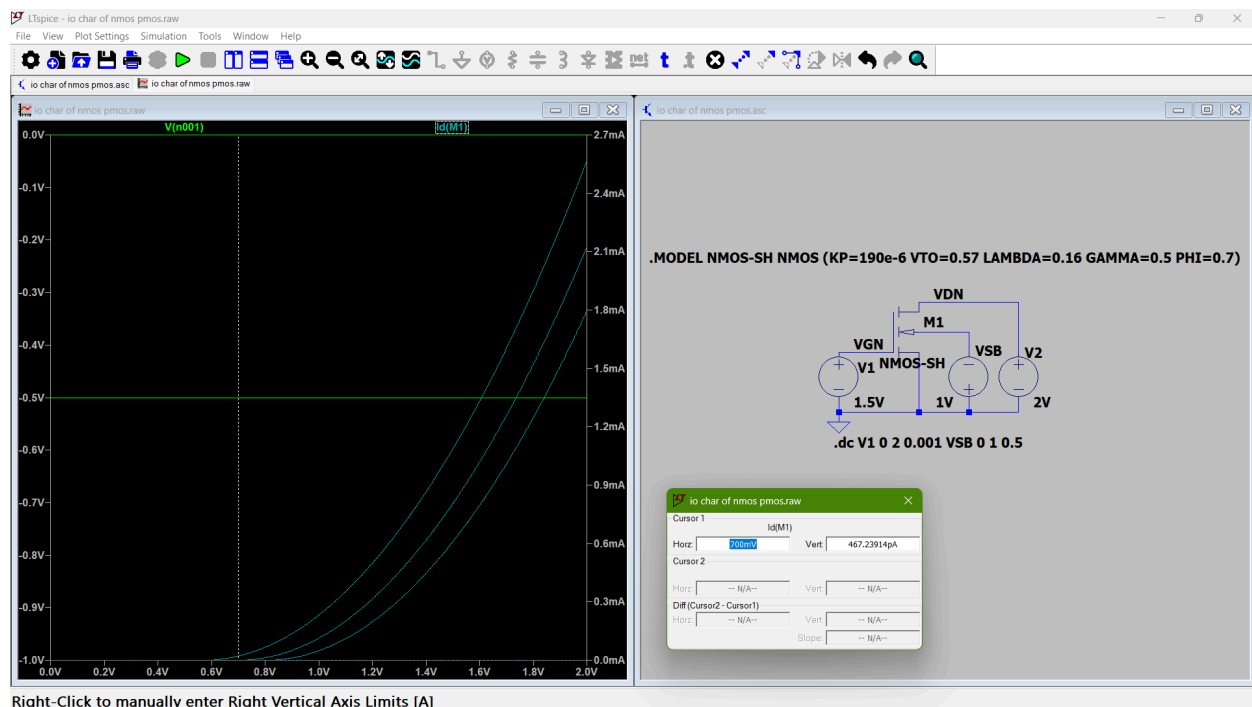
4. Simulation Results & Waveform Analysis

The nested simulation sweeps three overlaid traces across the screen.

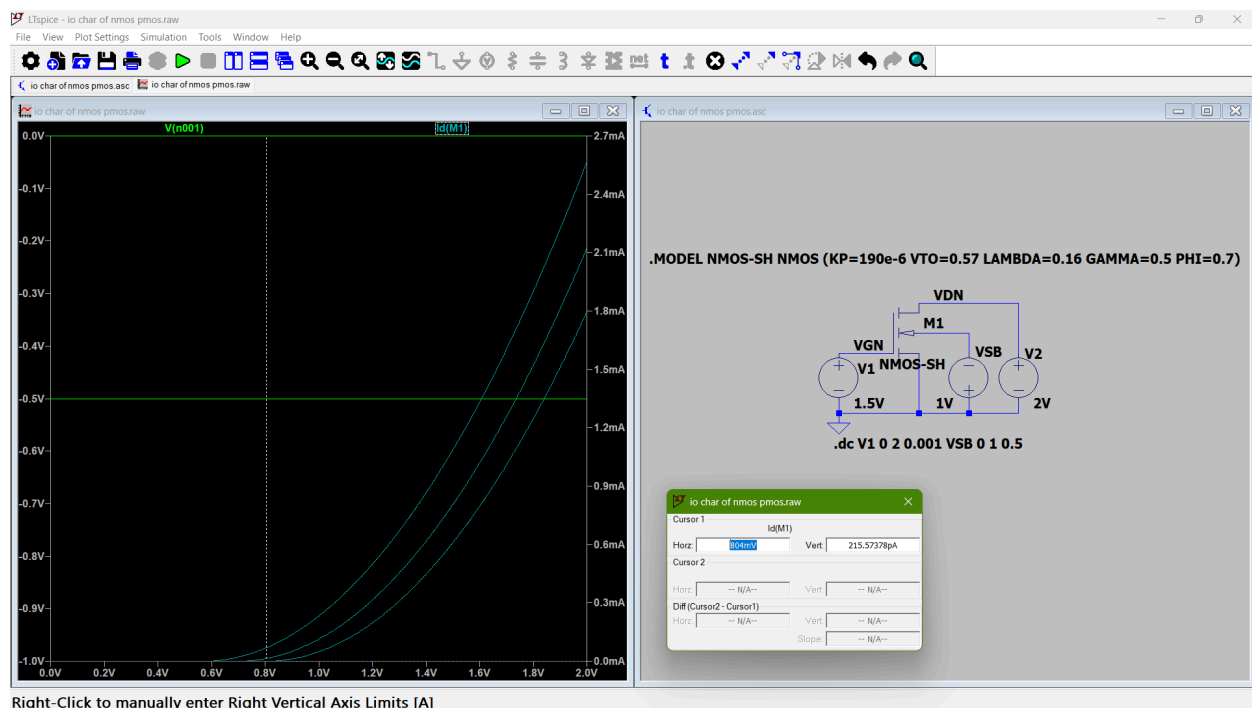
For $V_{SB} = 0V$:



For $V_{SB} = 0.5V$:



For $V_{SB} = 1V$:



A. Core Qualitative Waveform Behavior

The visual family of curves exhibits a distinct, parallel translation to the right along the horizontal axis as V_{SB} steps from $0V \rightarrow 0.5V \rightarrow 1V$. This rightward translation proves

that as the substrate reverse-bias increases, a significantly higher gate potential (V_{GS}) is strictly required to activate the device and initiate conduction.

B. Extraction of Simulated Threshold Voltages

By deploying tracking cursors to find the exact point where each independent current trace departs from the 0A baseline and begins its quadratic saturation ascension, the simulated threshold parameters were extracted:

- **Trace 1 ($V_{SB} = 0V$):** Turns on at $V_{GS} = 571\text{ mV}$
- **Trace 2 ($V_{SB} = 0.5V$):** Turns on at $V_{GS} = 700\text{ mV}$
- **Trace 3 ($V_{SB} = 1V$):** Turns on at $V_{GS} = 804\text{ mV}$

5. Quantitative Verification (Theory vs. Simulation)

The extracted threshold parameters from the live simulation graph are mapped directly against our analytical calculations below:

Reverse Body Bias (V_{SB})	Theoretical Threshold (V_{th})	Simulated Turn-on Point	Absolute Error (%)
0.0 V	0.5700 V	0.5710 V	0.175%
0.5 V	0.6994 V	0.7000 V	0.085%
1.0 V	0.8036 V	0.8040 V	0.050%

Validation Statement

The maximum absolute error among all evaluated operating profiles is a negligible 0.175%. This tight, sub-percent alignment confirms that the simulator is precisely executing the non-linear square-root body effect math defined by the $\gamma=0.5$ parameter.

6. Comprehensive Engineering Conclusion (The Solid-State Physics)

The data compiled across this investigation demonstrates the significant impact of substrate biasing on IC behavior. When a negative potential is applied to the bulk substrate relative to the channel source ($V_{SB} > 0$), it widens the depletion region beneath the gate oxide layer. This expansion exposes a greater concentration of immobile, negatively charged acceptor ions within the p-type substrate.

Because these extra negative bulk charges contribute to the local electric field, they block the vertical field arriving from the gate. Consequently, the gate voltage must work harder to neutralize these extra bulk charges before it can invert the surface channel and draw free electrons to form the inversion layer. This physical mechanism results in the threshold voltage scaling upward from 0.57V to 0.80V, an escalation directly dictated by the process parameter $\gamma=0.5$. In multi-tier analog layouts and digital logic sub-blocks sharing a common substrate, this threshold variance introduces significant propagation delays and gain degradation. Accounting for this variation is a critical constraint for robust, high-yield Analog IC design.